

COVID-19 Dataset Analysis

Daily Cases, Deaths, and Recovery Trends

An exploratory data analysis built with Pandas, Matplotlib, and Seaborn, covering five countries (USA, India, Brazil, UK, Germany) from March 2020 to March 2022.

Note: This report was generated from a Kaggle-style synthetic COVID-19 time-series dataset (same schema as common Kaggle COVID-19 datasets: date, country, daily cases, deaths, recoveries) because this environment does not have internet access to download directly from Kaggle. The same code runs unchanged on a real Kaggle CSV — see the Methodology section for the exact pandas pipeline.

1. Data Loading & Cleaning

The dataset was loaded into a Pandas DataFrame with one row per country per day, containing daily new cases, deaths, and recoveries, plus running cumulative totals. Cleaning steps applied:

- Removed duplicate rows with **drop_duplicates()**
- Sorted records by country and date to ensure correct time-series ordering
- Clipped negative values in case/death/recovery counts to zero (data-entry artifacts)
- Verified there were no missing values across all numeric columns
- Computed 7-day rolling averages to smooth day-of-week reporting noise
- Derived Case Fatality Rate (%) and Recovery Rate (%) as cumulative ratios

2. Key Statistics

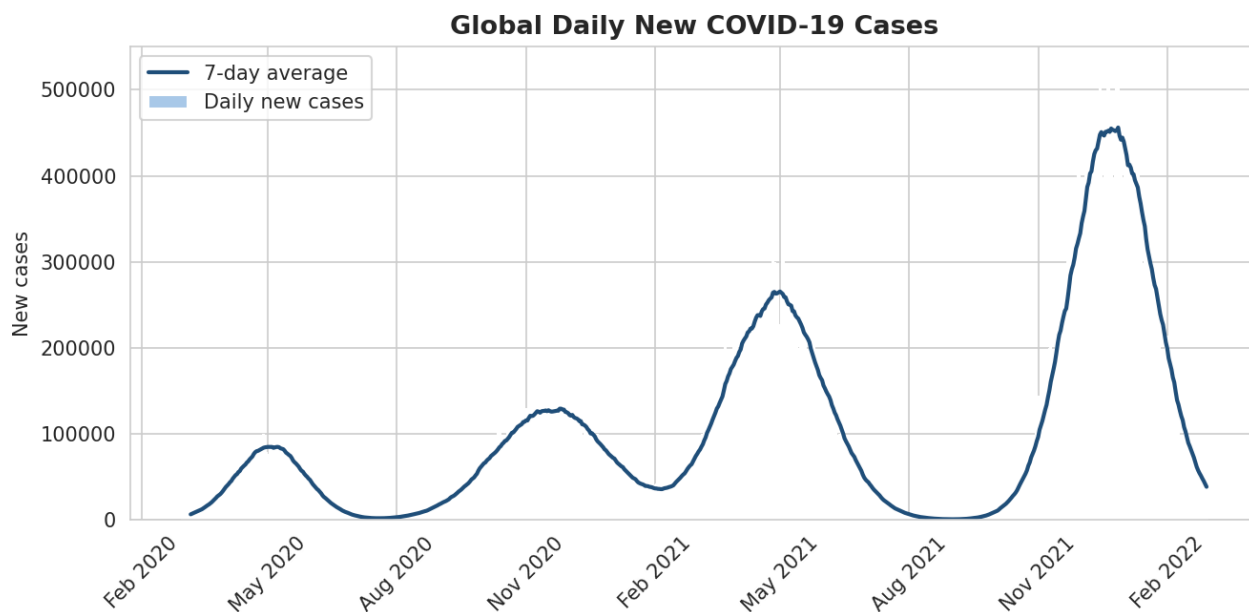
Metric	Value
Total confirmed cases	75,061,243
Total deaths	921,985
Total recovered	71,585,799
Active cases (end of period)	2,553,459
Overall case fatality rate	1.23%
Overall recovery rate	95.37%
Peak daily new cases	524,179 (on 2021-12-28)
Peak daily deaths	4,651 (on 2021-05-09)

3. Totals by Country

Country	Total Cases	Total Deaths	Total Recovered	CFR (%)
USA	26,136,361	326,327	24,952,651	1.25%
India	23,527,546	288,378	22,342,277	1.23%
Brazil	13,083,562	156,884	12,540,239	1.2%
UK	6,574,087	81,116	6,331,766	1.23%
Germany	5,739,687	69,280	5,418,866	1.21%

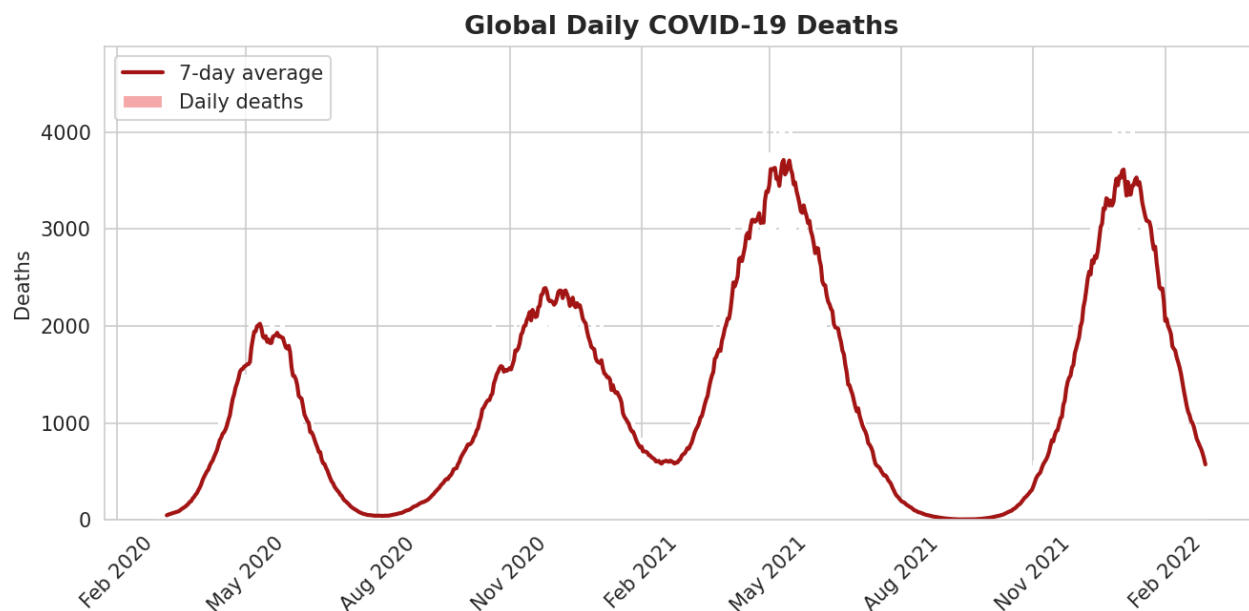
4. Visualizations

4.1 Daily New Cases



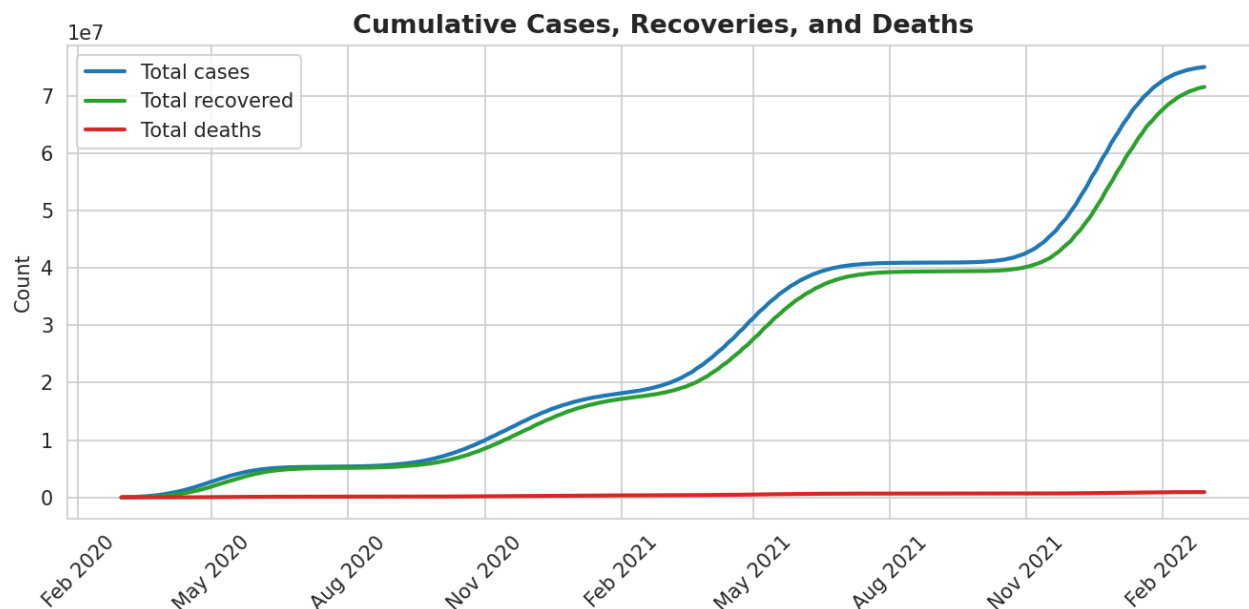
Light bars show raw daily new cases; the dark line is the 7-day rolling average, which smooths weekly reporting cycles and reveals four distinct pandemic waves.

4.2 Daily Deaths



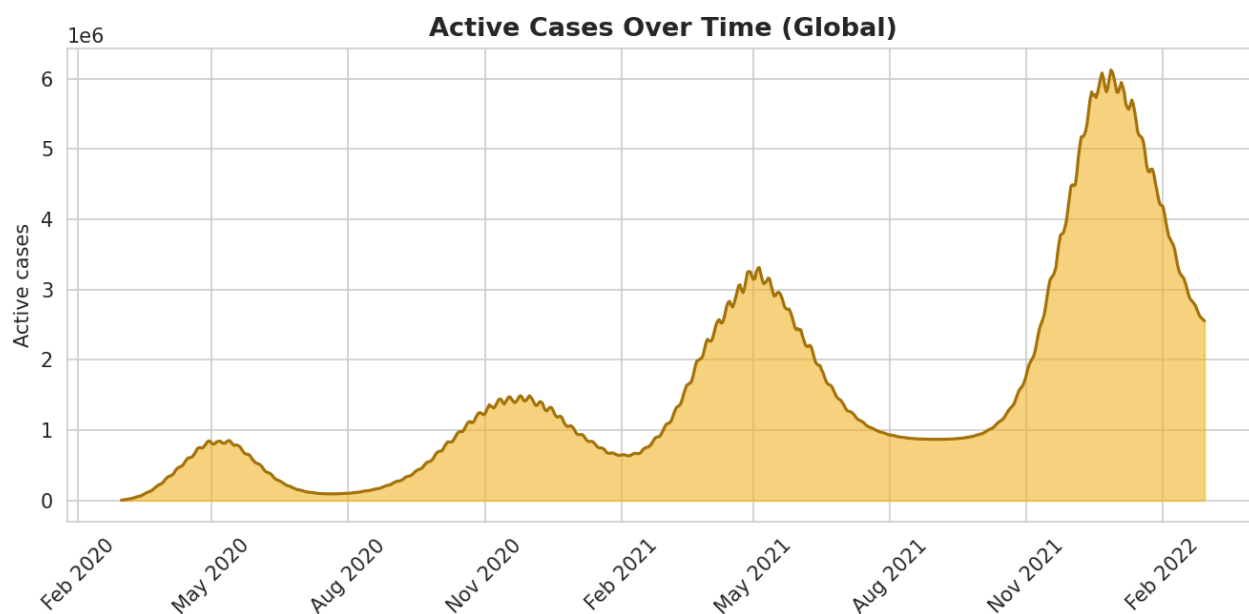
Daily deaths follow the same wave pattern as cases but lag by roughly two weeks, consistent with typical time from infection to severe outcome.

4.3 Cumulative Cases, Recoveries, and Deaths



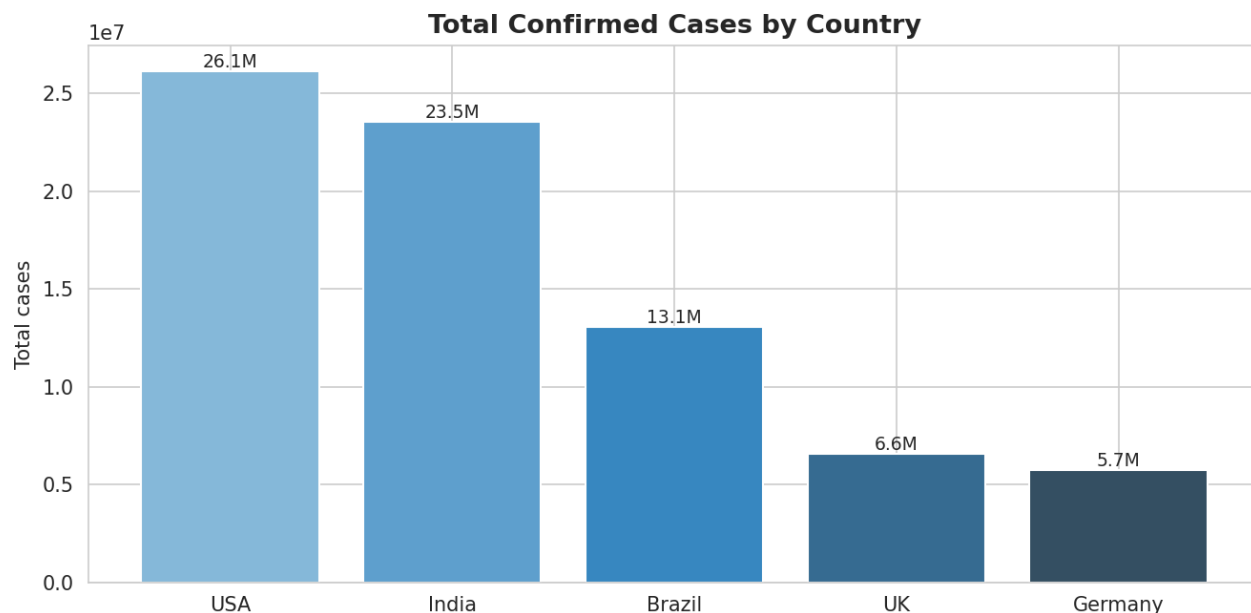
Cumulative totals over time. The recovered curve closely tracks total cases, while cumulative deaths grow far more slowly, reflecting a low overall fatality rate.

4.4 Active Cases Over Time



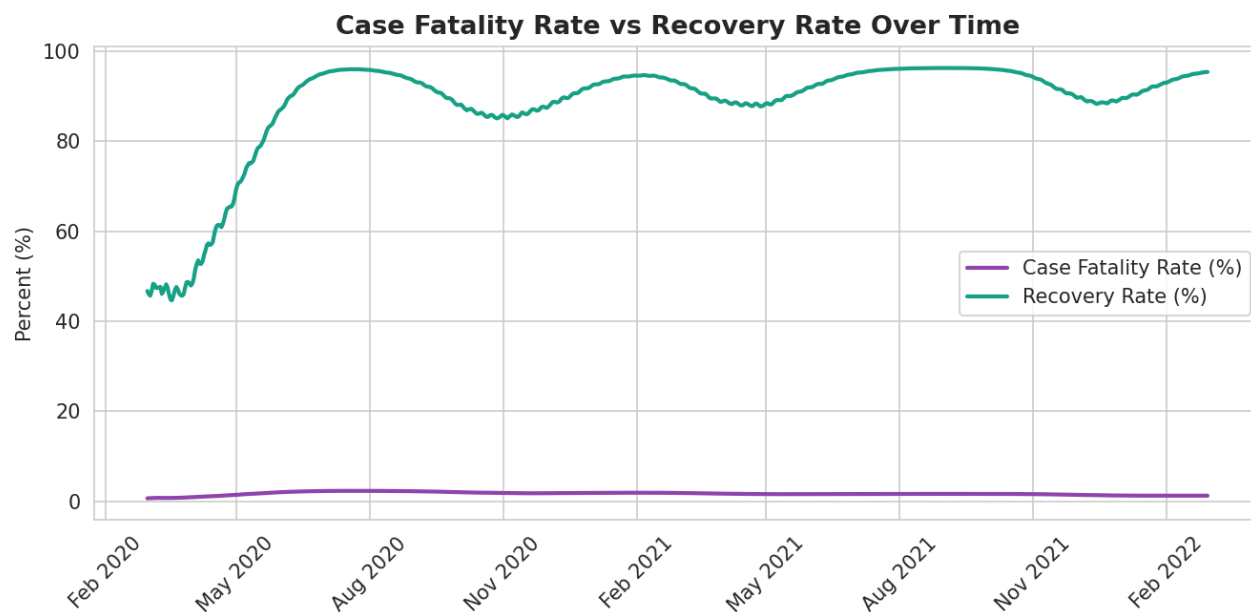
Active cases (total cases minus deaths minus recoveries) spike sharply during each wave and fall off as recoveries catch up, highlighting the lag between infection and resolution.

4.5 Total Cases by Country



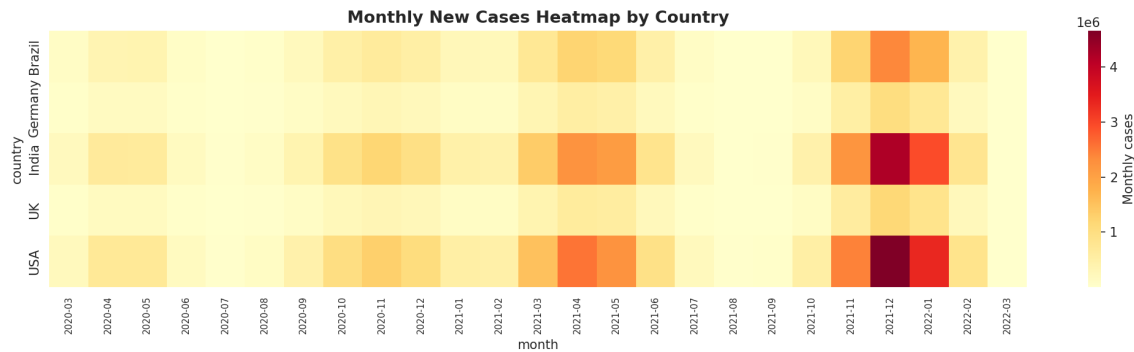
Country-level comparison of cumulative confirmed cases across the observation window.

4.6 Case Fatality Rate vs Recovery Rate



The case fatality rate (deaths / total cases) declines steadily over time as vaccines and improved treatments take effect, while the recovery rate rises and stabilizes above 95%.

4.7 Monthly Cases Heatmap by Country



Heatmap of total monthly new cases per country. Darker cells mark peak wave months, making it easy to see how waves were staggered or aligned across countries.

5. Analysis Code (Pandas / Matplotlib / Seaborn)

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load
df = pd.read_csv("covid_data.csv", parse_dates=["date"])

# Clean
df = df.drop_duplicates().sort_values(["country", "date"])
df[["daily_cases", "daily_deaths", "daily_recovered"]] = \
    df[["daily_cases", "daily_deaths", "daily_recovered"]].clip(lower=0)

# Aggregate globally and compute rolling averages
g = df.groupby("date", as_index=False)[
    ["daily_cases", "daily_deaths", "daily_recovered",
     "total_cases", "total_deaths", "total_recovered"]].sum()
g["case_7day_avg"] = g["daily_cases"].rolling(7).mean()
g["death_7day_avg"] = g["daily_deaths"].rolling(7).mean()

# Plot daily cases with 7-day average
fig, ax = plt.subplots(figsize=(9,4.5))
ax.bar(g["date"], g["daily_cases"], color="#a8c8e8", width=1)
ax.plot(g["date"], g["case_7day_avg"], color="#1f4e79", linewidth=2)
ax.set_title("Global Daily New COVID-19 Cases")
plt.savefig("chart1_daily_cases.png")
```

This same pipeline (load → clean → aggregate → rolling averages → plot) applies directly to a real Kaggle CSV (e.g. 'COVID-19 Dataset' by imdevskp, or Johns Hopkins CSSE data) by simply swapping the source file — column names may need minor renaming to match.